

Lecture 19: Numerical Computing

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1 Introduction

We employ numerical algorithm to compute the price of an American option in this lecture. We assume the classical Black-Scholes model:

$$dS_t = rS_t dt + \sigma S_t dW_t, \quad S_0 > 0, \quad (1)$$

where $r, \sigma > 0$ are respectively the risk-free rate and volatility. Our objective is to price numerically an American put with strike K and maturity T .

Last time we discussed the theoretical framework of pricing. We know that the put option price is determined by the “free boundary” backward Black-Scholes PDE:

$$\frac{\partial P}{\partial t} + rx \frac{\partial P}{\partial x} + \frac{1}{2} \sigma^2 x^2 \frac{\partial^2 P}{\partial x^2} = rP, \text{ for any } (t, x) \text{ s.t. } P(t, x) > (K - x)^+. \quad (2)$$

$$P(T, x) = (K - x)^+, \quad (3)$$

$$\lim_{x \rightarrow \infty} P(t, x) = 0. \quad (4)$$

The additional condition in (2) make the American characteristic appear. In fact, the set in which $P(t, x) > K - x$ determines the continuation region $\mathcal{C}$. If we ignore this additional condition than (2) and (3) just give us the price of an European put. European put price is computed with backward recursion, or conditional expectations. For American derivatives, we not only need conditional expectation but also need exercise checking: we need to compare the discounted conditional expectation with the intrinsic value at that time, and take the maximum of the two be the price of the put at that time. This suggests two differential numerical algorithms for pricing, which will be discussed separately in what follows.

2 Finite difference

Finite difference is a very useful method to solve PDE numerically. The basic idea is to discretize both the time space and spacial space. In particular, we need to rewrite (2) into a difference equation in order to employ finite difference. Let us respectively denote by $\delta t > 0$ and $\delta x > 0$ the time step and spacial step, then from (2) we have a finite difference equation that is satisfied by $P(t, x)$:

$$P(t + \delta t, x) - P(t, x) + \left(rx \frac{P(t, x + \delta x) - P(t, x - \delta x)}{2(\delta x)} + \frac{1}{2} \sigma^2 x^2 \frac{P(t, x + \delta x) - 2P(t, x) + P(t, x - \delta x)}{(\delta x)^2} - rP(t, x) \right) \delta t = 0. \quad (5)$$

The step sizes δt and δx are determined by the total number of time steps and the total number of space steps. In order to compute the price at time t when the stock price is x , let us define

$$\delta t = \frac{T - t}{M}, \quad \delta x = \frac{2x}{N}, \quad (6)$$

where M is the number of time steps from time t to time T , and N is an even number, representing the number of space steps from δx to $2x + \delta x$. To compute $P(t, x)$, we would like to determine the put price on all grid points $(t + i \cdot (\delta t), j \cdot (\delta x))$ for $0 \leq i \leq M$ and $1 \leq j \leq N + 1$. And this computation is done backwardly in time from $i = M$ to $i = 0$.

Let us denote by $t_i = t + i \cdot (\delta t)$, $x_j = j \cdot (\delta x)$. Then we can rewrite (5) as

$$P(t_{i+1}, x_j) - P(t_i, x_j) + \left(r \cdot j \frac{P(t_i, x_{j+1}) - P(t_i, x_{j-1})}{2} + \frac{1}{2} \sigma^2 \cdot j^2 [P(t_i, x_{j+1}) - 2P(t_i, x_j) + P(t_i, x_{j-1})] - rP(t_i, x_j) \right) \delta t = 0, \quad (7)$$

from which we obtain that

$$P(t_{i+1}, x_j) = 0.5(-r \cdot j - \sigma^2 \cdot j^2) \delta t P(t_i, x_{j+1}) + (1 + \sigma^2 \cdot j^2 \delta t + r \delta t) P(t_i, x_j) + 0.5(r \cdot j - \sigma^2 \cdot j^2) \delta t P(t_i, x_{j-1}) \quad (8)$$

We can re-express (8) in terms of vectors and matrix. In particular, let us denote by

$$\mathbb{A} = \begin{pmatrix} 1 + (\sigma^2 + r) \delta t & 0.5(-r - \sigma^2) \delta t & 0 & 0 & \dots & 0 \\ 0.5(2r - 4\sigma^2) \delta t & 1 + (4\sigma^2 + r) \delta t & 0.5(-2r - 4\sigma^2) \delta t & 0 & \dots & 0 \\ 0 & 0.5(3r - 9\sigma^2) \delta t & 1 + (9\sigma^2 + r) \delta t & 0.5(-3r - 9\sigma^2) \delta t & \dots & 0 \\ 0 & 0 & 0.5(4r - 16\sigma^2) \delta t & 1 + (16\sigma^2 + r) \delta t & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{pmatrix}, \quad (9)$$

$$\mathbb{P}_i = (P(t_i, x_1) \quad \dots \quad P(t_i, x_j) \quad \dots \quad P(t_i, x_{N+1}))^T. \quad (10)$$

Then we have a recursive system

$$\mathbb{A} \cdot \mathbb{P}_i = \mathbb{P}_{i+1}, \quad (11)$$

and terminal condition

$$\mathbb{P}_M = P(T, x_j) = (K - x_j)^+. \quad (12)$$

From (12) we have

$$\mathbb{P}_i = \mathbb{A}^{-1} \cdot \mathbb{P}_{i+1}, \quad (13)$$

so that we can use backward recursion to get the vector $\mathbb{P}_{M-1}$. What is the meaning of $\mathbb{P}_{M-1}$? One can interpret $\mathbb{P}_{M-1}$ as the discounted expectation of $\mathbb{P}_M$. This is because we obtain $\mathbb{P}_{M-1}$ from $\mathbb{P}_M$ by “solving” the B-S PDE. To get the real put price at time $t_{M-1} = T - \delta t$, we need to check for exercise. To do so, we need to “process” the vector $\mathbb{P}_{M-1}$: we have the vector produced by (14)

$$\mathbb{P}_{M-1} = (P(t_{M-1}, x_1), \dots, P(t_{M-1}, x_j), \dots, P(t_{M-1}, x_{N+1}))^T. \quad (14)$$

Then we re-define the components of $\mathbb{P}_{M-1}$:

$$P(t_{M-1}, x_j) = \max\{(K - x_j)^+, P(t_{M-1}, x_j)\}. \quad (15)$$

These new components give us the put price at time $t_{M-1} = T - \delta t$. By induction, we get the put prices at time $t_0 = t$. And the quantity of interest is given by

$$P(t, x) = P(t_0, x_{\frac{N}{2}}). \quad (16)$$

Example 1. Using the Matlab code on course website we can compute the put price

$$P(0, 50) = 3.3078, \quad (17)$$

where the spot price is 50, the strike is 50, the interest rate is 0.05, the volatility is 0.2, maturity is 1, both the number of time steps and space steps are 200.

Remark 1. If we ignore the step in (15) then we obtain the European put price $P(t, x)$, which should be smaller than its American counterpart. This European put price given by finite difference will also be slightly smaller than the one given by B-S formula. This may be due to discretization.

Remark 2. Because of the fact that

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}, \quad (18)$$

one may want to choose a very large M and N , or equivalently, very small δt and δx to implement the above finite difference scheme. The fact is, it is not practically useful or is even inappropriate. In practice, computer usually has a limited precision, which is the smallest positive number the computer can understand. For example, if $\epsilon = 10^{-6}$ is the precision, then this computer cannot distinguish 0 from 10^{-7} . So for any function f defined around x

$$\frac{f(x + 10^{-7}) - f(x)}{10^{-7}} = \frac{0}{10^{-7}} = 0. \quad (19)$$

How can we effectively approximate the first derivative? Let us illustrate this using $\frac{\partial P}{\partial t}(t, x)$: we search for an appropriate $h > 0$ to compute this derivative. Using Taylor's expansion, we have

$$\frac{P(t+h, x) - P(t, x)}{h} = \frac{p_t(t, x)h + 0.5P_{tt}(\xi, x)h^2 + \epsilon}{h}, \quad (20)$$

where $\xi \in (t, t+h)$ and ϵ appears because of the precision. From (19) we see that the error of estimating $P_t(t, x)$ is given by

$$\frac{1}{2}P_{tt}(\xi, x)h + \frac{\epsilon}{h}. \quad (21)$$

For a good function we can assume that $P_{tt}(\xi, x)$ is a bounded number, so we should choose $h = O(\sqrt{\epsilon})$. That is, if $\epsilon = 10^{-6}$, we'd better let $h > 10^{-3}$.

3 Tree

Binomial tree can also be used to price an American option. We do not repeat the algorithm but will just state the model setup.

To price an American put for the process in (1), we consider the following N -period binomial model: assume the current state is (t, x) , then we set

- Time step $\delta t = (T - t)/N$;
- One-period interest rate $R = e^{r\delta t} - 1$;
- Up factor $u = e^{\sigma\sqrt{\delta t}}$ and down factor $d = 1/u$;
- Risk-neutral probabilities $\tilde{p} = \frac{1+R-d}{u-d}$ and $\tilde{q} = 1 - \tilde{p}$.